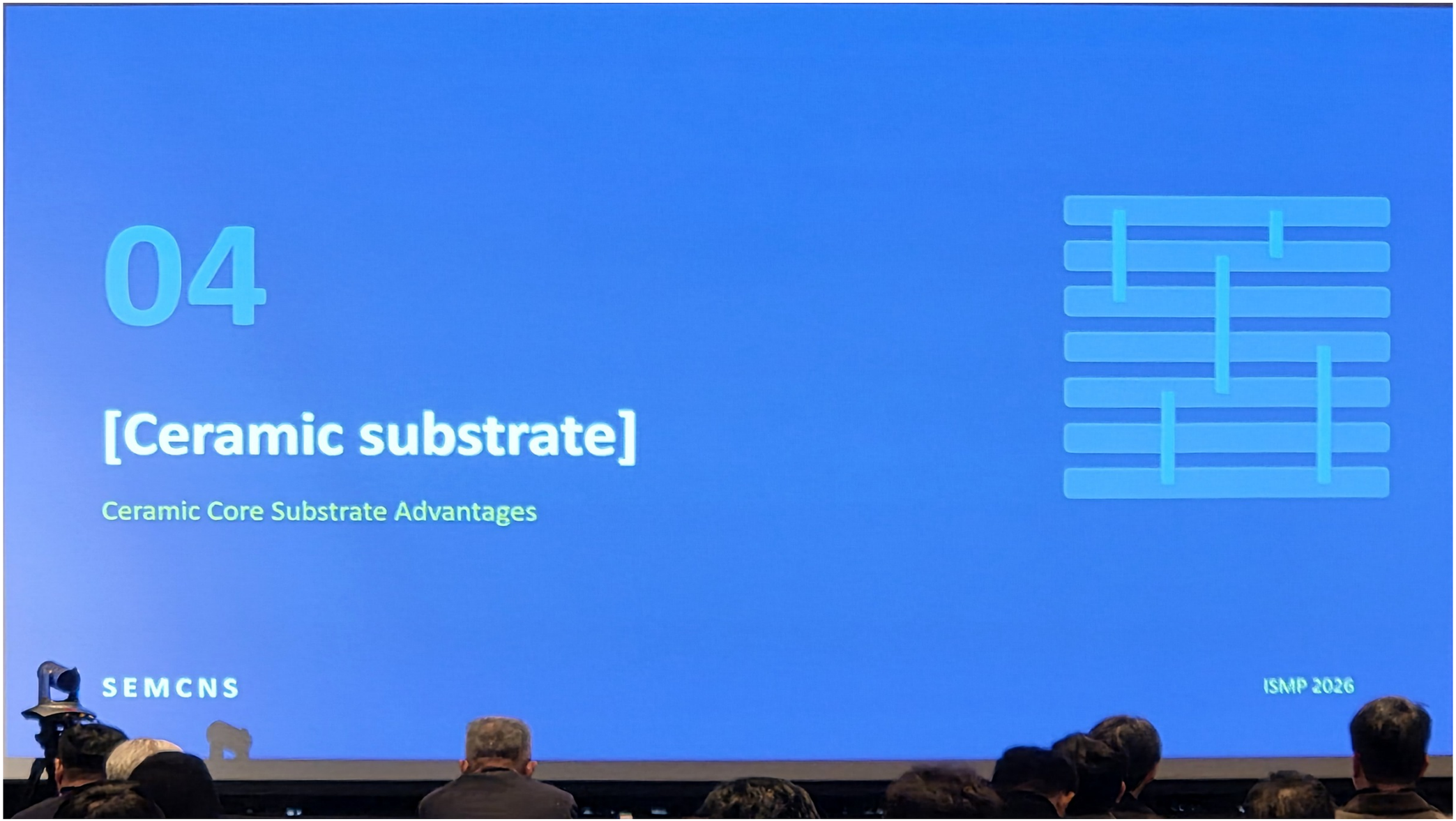
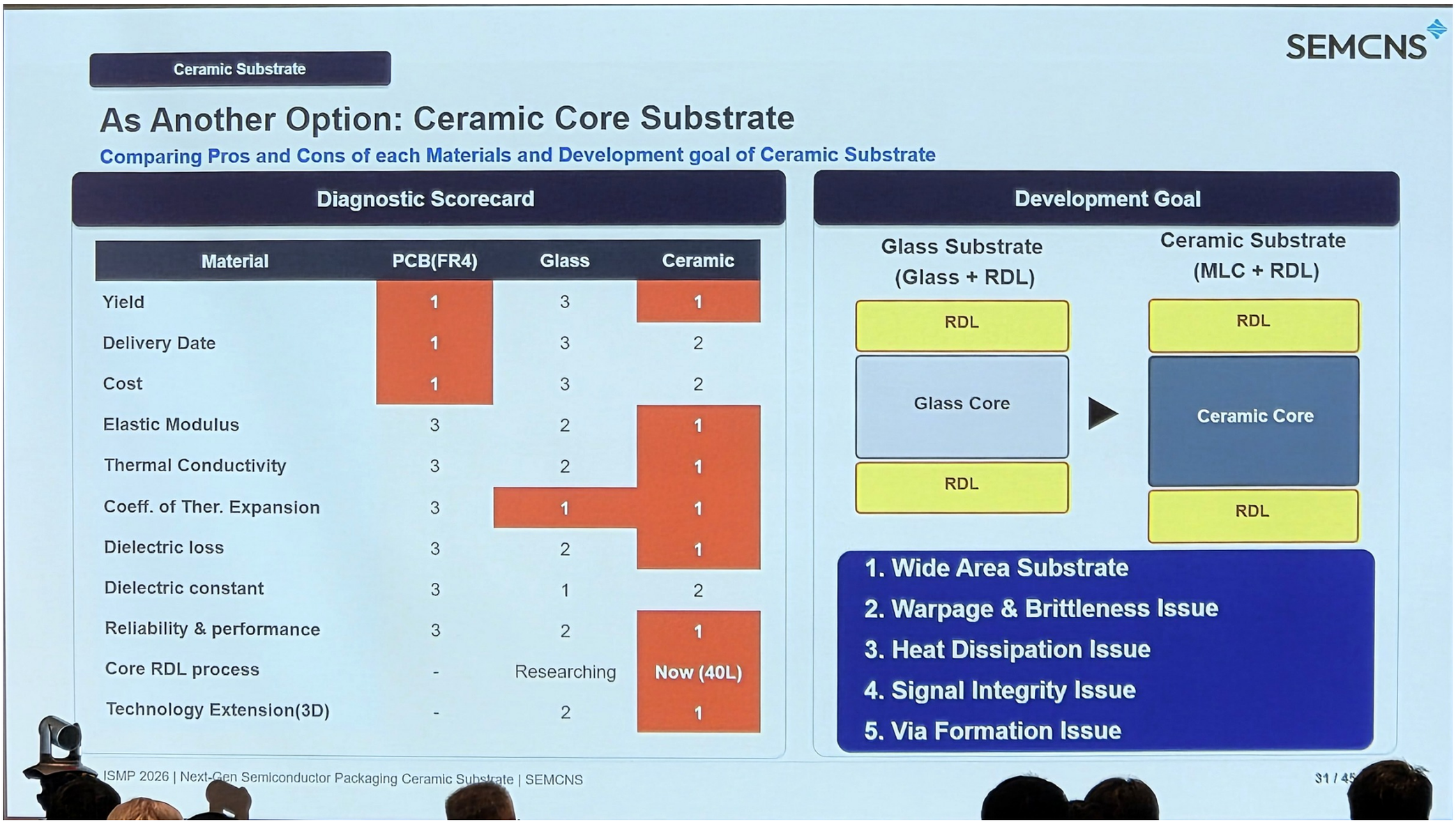
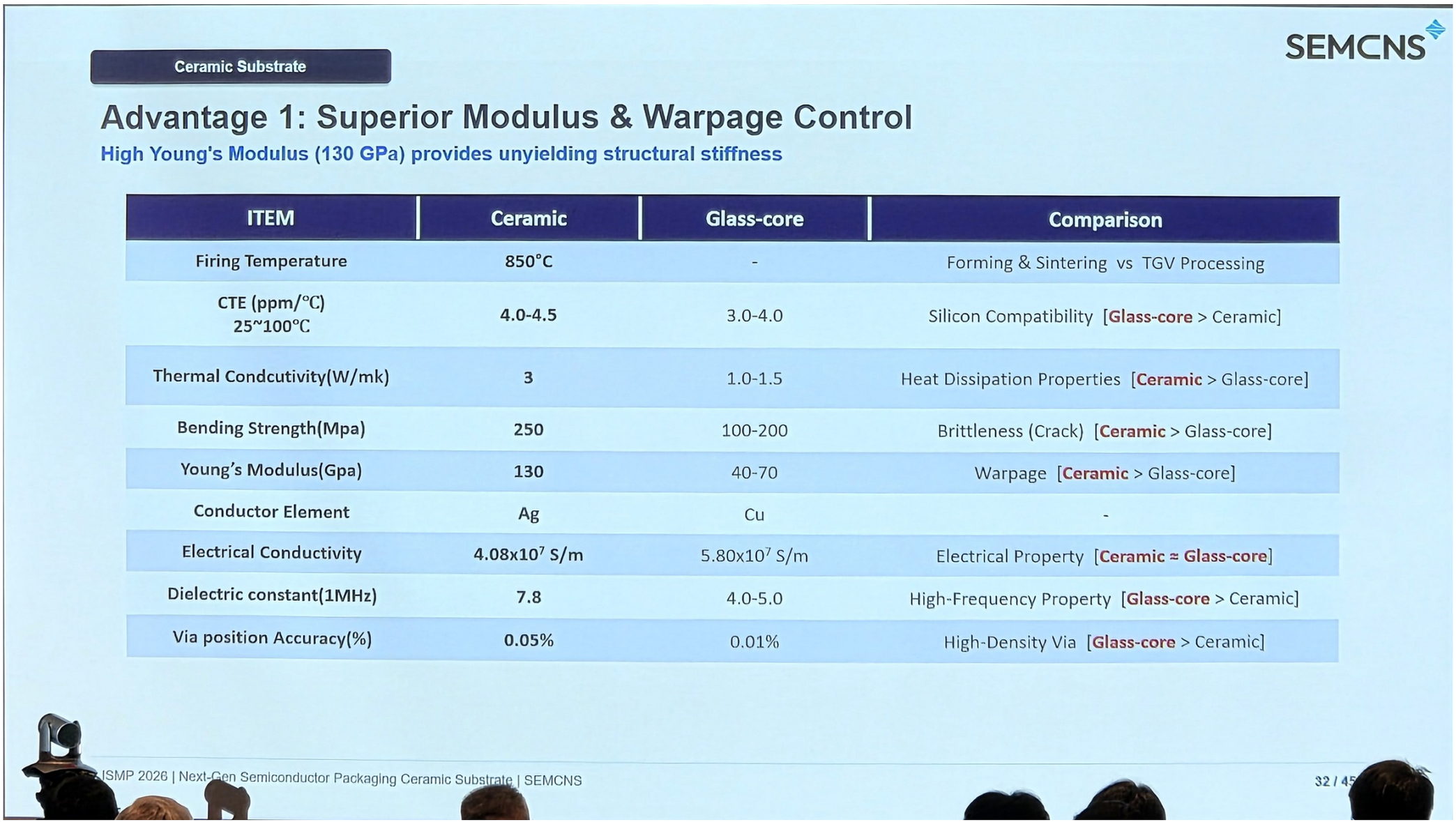
T35 세라믹 코어 기판의 휨·방열·비아 공정

SEMCNS · Dae-Hyeong Lee · 이대형



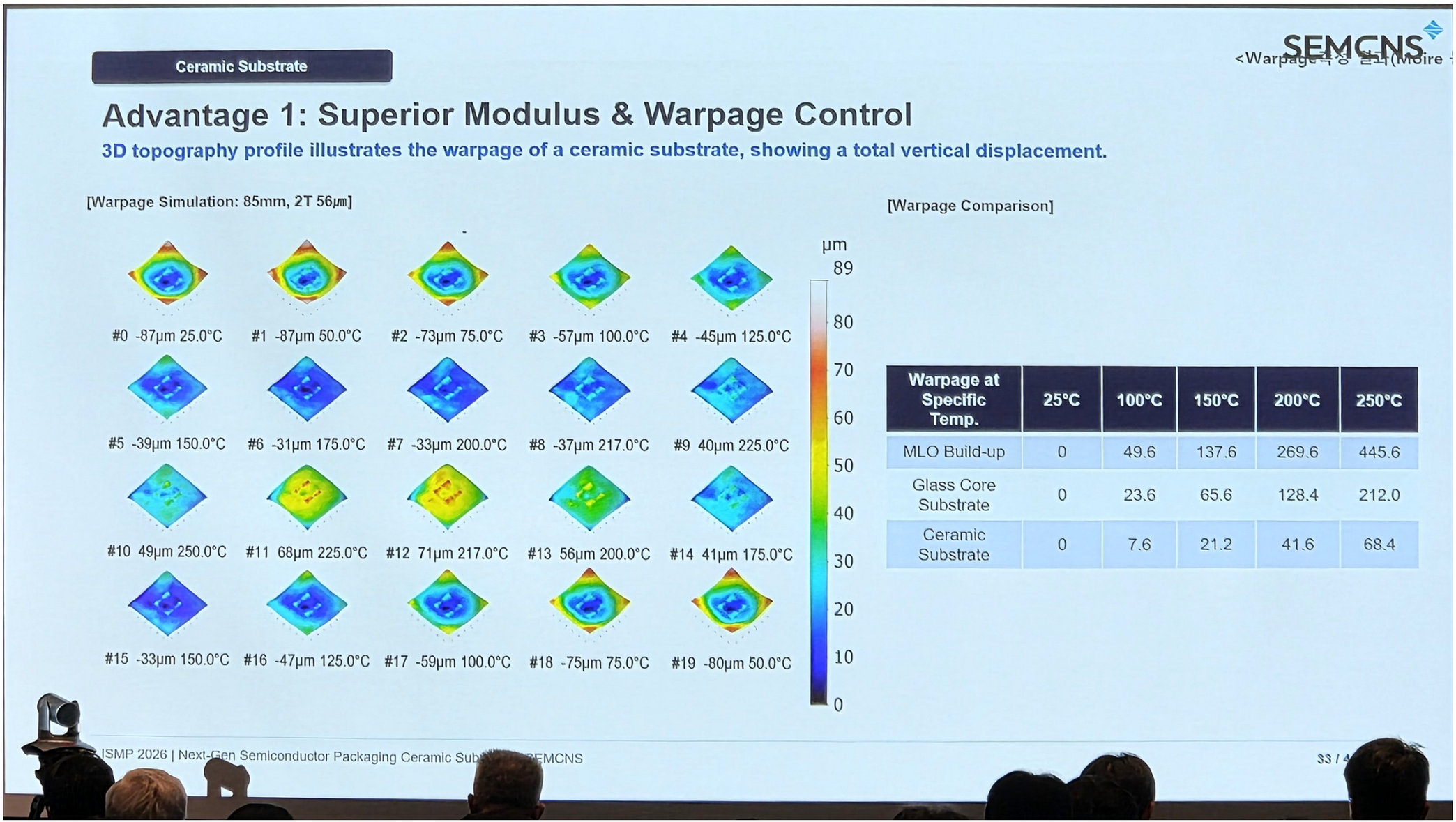

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Ceramic Substrate
As Another Option: Ceramic Core Substrate
Comparing Pros and Cons of each Materials and Development goal of Ceramic Substrate
Diagnostic Scorecard
Material | PCB(FR4) | Glass | Ceramic
Yield | 1 | 3 | 1
Delivery Date | 1 | 3 | 2
Cost | 1 | 3 | 2
Elastic Modulus | 3 | 2 | 1
Thermal Conductivity | 3 | 2 | 1
Coeff. of Ther. Expansion | 3 | 1 | 1
Dielectric loss | 3 | 2 | 1
Dielectric constant | 3 | 1 | 2
Reliability & performance | 3 | 2 | 1
Core RDL process | - | Researching | Now (40L)
Technology Extension(3D) | - | 2 | 1
Development Goal
Glass Substrate (Glass + RDL)
RDL
Glass Core
RDL
Ceramic Substrate (MLC + RDL)
RDL
Ceramic Core
RDL
1. Wide Area Substrate
2. Warpage & Brittleness Issue
3. Heat Dissipation Issue
4. Signal Integrity Issue
5. Via Formation Issue
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ITEM	Ceramic	Glass-core	Comparison
Firing Temperature	850°C	-	Forming & Sintering vs TGV Processing
CTE (ppm/℃) 25~100℃	4.0-4.5	3.0-4.0	Silicon Compatibility [Glass-core > Ceramic]
Thermal Condcutivity(W/mk)	3	1.0-1.5	Heat Dissipation Properties [Ceramic > Glass-core]
Bending Strength(Mpa)	250	100-200	Brittleness (Crack) [Ceramic > Glass-core]
Young's Modulus(Gpa)	130	40-70	Warpage [Ceramic > Glass-core]
Conductor Element	Ag	Cu	-
Electrical Conductivity	$4.08x10^7$ S/m	$5.80x10^7$ S/m	Electrical Property [Ceramic ≈ Glass-core]
Dielectric constant(1MHz)	7.8	4.0-5.0	High-Frequency Property [Glass-core > Ceramic]
Via position Accuracy(%)	0.05%	0.01%	High-Density Via [Glass-core > Ceramic]

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Warpage at Specific Temp.	25°C	100°C	150°C	200°C	250°C
MLO Build-up	0	49.6	137.6	269.6	445.6
Glass Core Substrate	0	23.6	65.6	128.4	212.0
Ceramic Substrate	0	7.6	21.2	41.6	68.4

Ceramic Substrate

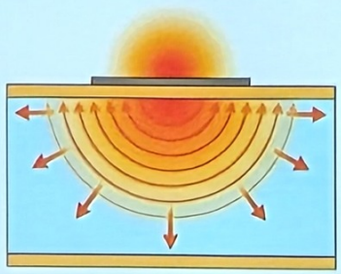
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Advantage 2: High Thermal Conductivity & Heat Spreading

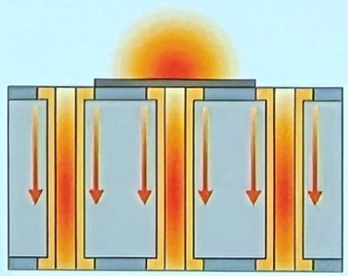
Superior thermal conductivity (2.4~3.0 W/m·k) dissipates high TDP(Thermal Design Power) heat efficiently

THERMAL METRICS & DISSIPATION MECHANISMS

SUBSTRATE	THERMAL CONDUCTIVITY (W/mK)
MLO	<0.5 W/mK
Glass	1.1 W/mK
Ceramic	**2.4~3.0 W/mK - Excellent**

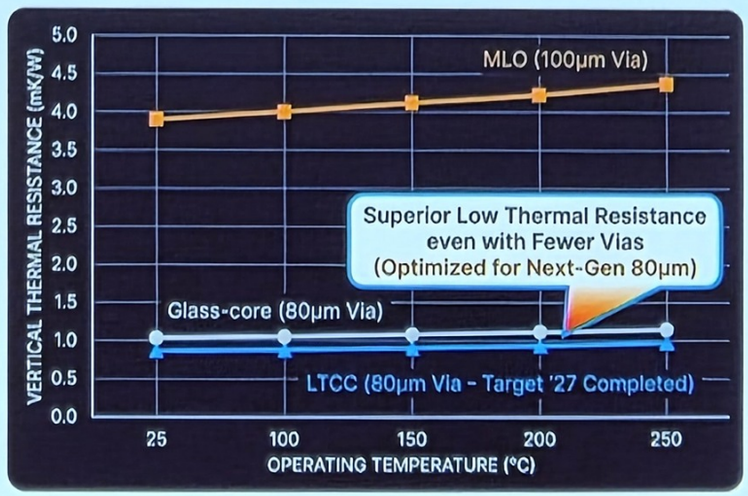


LTCC Ceramic Body:
Isotropic Heat Spreading

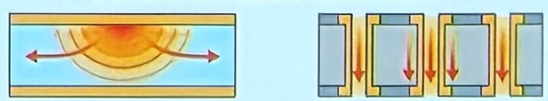


Glass-Core:
Vertical Heat Trapping

HIGH-TEMPERATURE VIA THERMAL PERFORMANCE (25°C - 250°C)



VIA HEAT DIFFUSION MECHANISM



1. LTCC (80µm Via)

2. Glass-core (80µm Via)

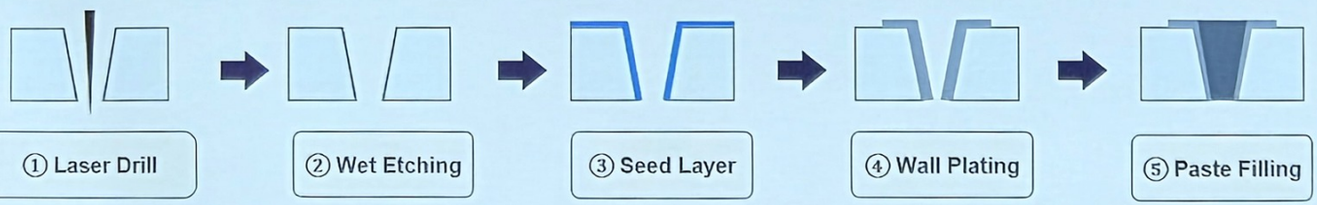
Ceramic Substrate

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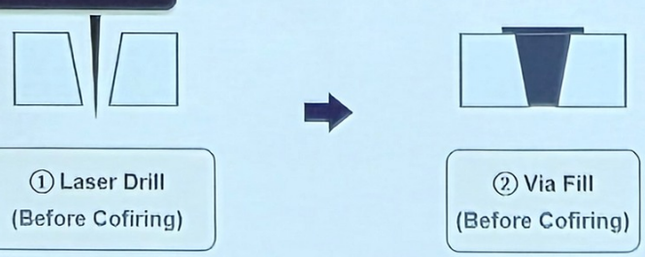
Advantage 3: Soft Via Green Sheet Machining Precision

Hole formation in flexible un-fired state eliminates thermal shock and micro-cracks

Glass Hard Via – Complex & Risky

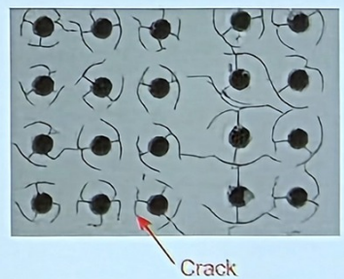


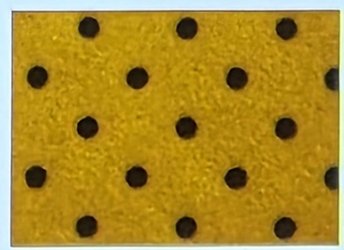
LTCC Soft Via – Simple & High Yield



A 2-step process optimized for "Cost Reduction" and "Production Scale-up"

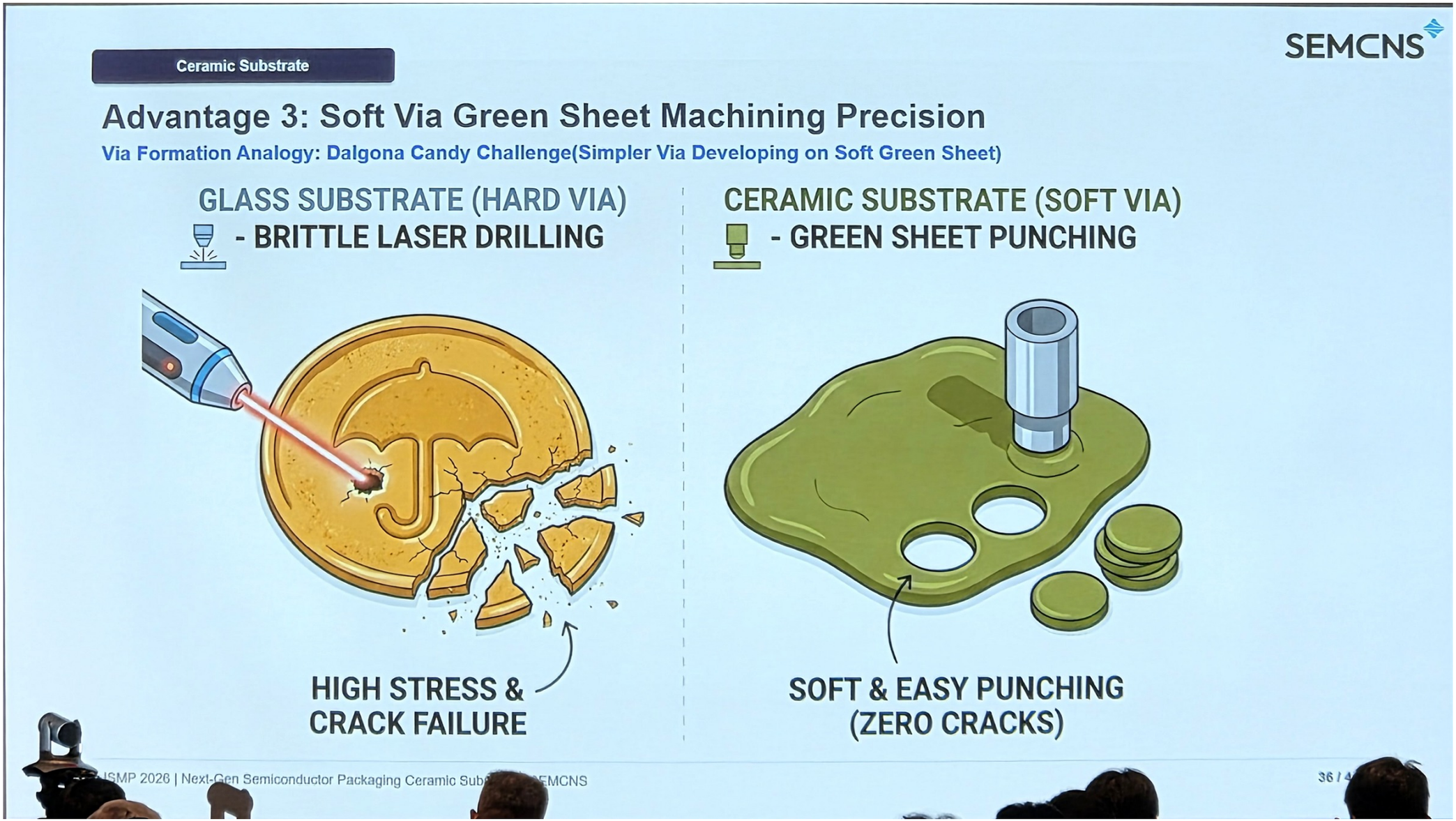
Via Comparison





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Ceramic Substrate
Advantage 3: Soft Via Green Sheet Machining Precision
Via Formation Analogy: Dalgona Candy Challenge(Simpler Via Developing on Soft Green Sheet)
GLASS SUBSTRATE (HARD VIA)
- BRITTLE LASER DRILLING
CERAMIC SUBSTRATE (SOFT VIA)
- GREEN SHEET PUNCHING
HIGH STRESS & CRACK FAILURE
SOFT & EASY PUNCHING (ZERO CRACKS)
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Ceramic Substrate

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Summary: Advantages of Ceramic Substrate

Solving mechanical, thermal, and manufacturing bottlenecks in AI packaging

PCB Substrate

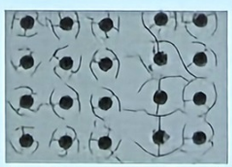
1. **High CTE mismatch & warpage**
2. Large-area fabrication limit
3. High electrical signal loss



Severe Warpage Defect

Glass Substrate

1. **Severe via crack issues**
2. No internal multi-layer routing
3. Thickness scaling constraints



Via Formation Cracks

Ceramic Substrate (Optional)

1. **Superior Warpage Control:** Low thermal deformation
2. **Large-Area Capability:** High mechanical rigidity
3. **High Integration:** Internal multi-layer circuit routing

Property	PCB	Glass	Ceramic
Thermal Cond. (W/m·K)	0.3	1.1	**2.6**
CTE (ppm/°C)	14~17	3.3	**4.0~4.4**
Elastic Modulus (GPa)	4.6	42	**130**
Bending Strength (MPa)	380	188	**280**

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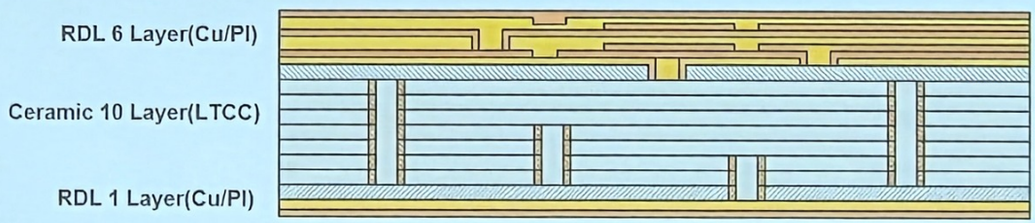
Development Status

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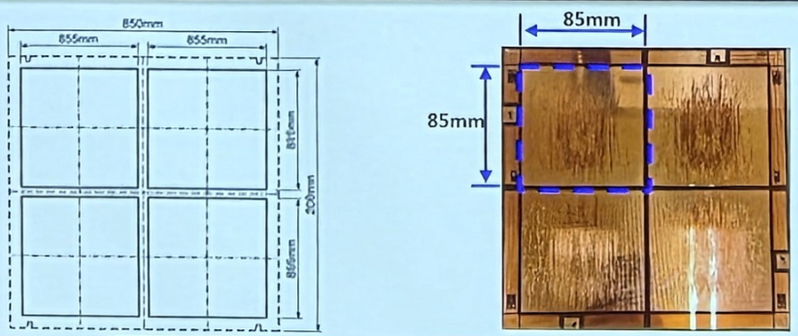
Production Samples & Exhibition Showcase

Large-area ceramic package substrate test Substrates

Substrate Stack-Up Structure

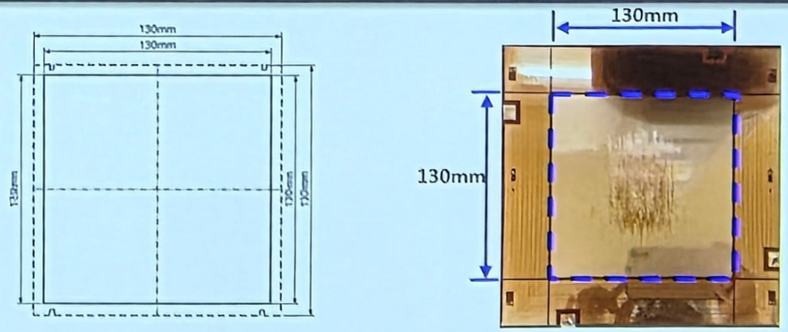


85X85mm 4-Unit Array



Structure & Electrical Test Complete

130mm 1-Unit



Warpage Test Complete

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Conclusion

Ceramic As an New Option for HPC

1. **Large-area stability**
   Low warpage and mechanical robustness
2. **Heat dissipation**
   Efficient heat removal for high-power packages
3. **Via accuracy**
   Green-sheet via formation with constrained sintering
4. **Proven in practice**
   Demonstrated with fabricated samples

